
NEXURANCE AI

Parametric Income Protection for India's Q-Commerce Workers

"We don't trust your location, we verify reality."

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1. The Problem

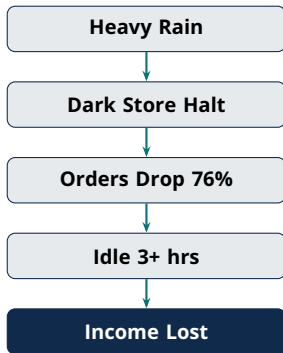
8M+ gig delivery workers. Zero income protection.

Ravi Kumar — Blinkit Rider, Bengaluru

- ▶ Heavy rain → Dark store halts → Orders drop **76%**
- ▶ Ravi idles **3+ hours** → Income lost
- ▶ Monthly loss from rain days: **Rs4,000–Rs6,000**

The Scale

- ▶ **85,000+** Q-commerce riders in Bengaluru alone
- ▶ **~8** disruption days per monsoon
- ▶ **Rs27–54 Cr** annual uninsured loss, Bengaluru
- ▶ Typical earnings: **Rs800–Rs1,200 / day**



2. Why Existing Solutions Fail

Traditional insurance wasn't built for Ravi's world.

Capability	Traditional Insurance	GPS-only Parametric	Nexurance AI
Claim Process	Manual filing	Single verification	✓ Zero-touch auto
Fraud Resistance	Investigator review	Easily spoofed	✓ 5-signal MSTs
Settlement Speed	7-30 days	Variable	✓ <60 seconds
Premium Model	Monthly	Variable	✓ Weekly Rs15-Rs60
Verification Layers	✗	GPS only	✓ 5 independent
Anti-Spoofing	✗	✗	✓ Fraud-resistant

Key Insight: GPS is not proof of presence — physical signals are.
We verify reality through **5 independent signal layers**: location consistency
· motion validity · network authenticity · behavioral history · cluster isolation

3. Nexurance AI — How It Works

Zero-touch parametric income protection. No claims. No paperwork. No waiting.

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5 Independent Signal Layers:

location consistency · motion validity · network authenticity · behavioral history · cluster isolation

4. Weekly Premium Model

Aligned to gig workers' weekly cash cycle. Dynamic. Actuarially viable.

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Premium Formula

$P = \text{Base} \times \text{Zone Risk} \times \text{Weather} \times \text{Trust Discount}$

Monsoon, HSR Layout: $\text{Rs}20 \times 1.3 \times 1.4 \times 0.90 = \text{Rs}33/\text{week}$

Dry week, Whitefield: $\text{Rs}20 \times 0.8 \times 1.0 \times 0.88 = \text{Rs}15/\text{week}$

Actuarial Viability

Avg. weekly premium	Rs32
Events/worker/monsoon month	1.8
Avg. payout per event	Rs480
Monsoon loss ratio	65%
Off-season loss ratio	18%

Blended annual: ~58% — within parametric benchmarks (55–70%). Viable without external subsidy.

Plan	Premium	Max Payout
Starter	Rs15–Rs25	Rs500
Shield	Rs26–Rs40	Rs1,200
Max	Rs41–Rs60	Rs2,500

58%

blended loss ratio
industry benchmark: 55–70%

5. Parametric Trigger Engine

Objective, third-party verified triggers only. No worker self-reporting.

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Level 1 — Environmental

Highest confidence · Auto-trigger

- ▶ Rainfall > 50mm/hr
(OpenWeatherMap + IMD)
- ▶ Flood alert Orange/Red
(IMD API)
- ▶ AQI > 400 hazardous
(CPCB API)
- ▶ Heat Index > 48°C
(IMD advisory)

Level 2 — Social / Civic

Med. confidence · Corroboration

- ▶ Government curfew
(notification APIs)
- ▶ Road closures > 70%
(traffic API)
- ▶ Dark stores paused > 60%
(platform API)

Level 3 — Platform Signals

Supporting evidence ONLY

- ▶ Order density drop > 40%
(simulated)
- ▶ Worker inactivity > 90 mins
peak hours

**Level 3 CANNOT alone
trigger a payout.
Eliminates platform-
manipulation attacks.**

500 workers. Fake GPS. Coordinated Telegram groups. Simple GPS is dead.

Trust = 0.25·Location + 0.25·Motion + 0.20·Network + 0.20·Behavior + 0.10·Cluster

Signal	How It Works
Location 25%	Cell tower + WiFi AP; GPS drift; spoofer cannot fake cellular BTS
Motion 25%	Accelerometer FFT; barometer drop; rain-specific sensor signature
Network 20%	VPN/proxy ASN detection; isMock() API; Play Integrity
Behavior 20%	Zone familiarity (90-day); claim baseline; earnings alignment
Cluster 10%	Louvain graph clustering; burst detection; same /24 subnet

0.80–1.00 TRUSTED

0.60–0.79 VERIFY

0.40–0.59 REVIEW

0.00–0.39 BLOCK

UX Balance: honest workers never penalised for network drops during verified rain (+0.08 forgiveness)

7. AI / ML Architecture

Three adaptive models — learn, retrain weekly, self-tighten under attack.

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Module 1 Dynamic Premium Engine

XGBoost Regression

8 features: zone risk, weather forecast, AQI, day-of-week, worker history, trust score

Retrains every Sunday night

Module 2 MSTS Fraud Detection

Isolation Forest anomaly detection

Louvain graph clustering
Temporal drift: 2.5σ deviation
auto-tightens threshold in real time

Module 3 Payout Eligibility Scorer

Logistic Regression

(interpretable — regulator ready)
Inputs: trigger strength, trust score, zone risk, time patterns

Output: Rs15–Rs60/week
weekly recalibration

Output: Trust score 0–1
adaptive thresholds

P > **0.80** → instant approve
P < **0.40** → blocked

Adaptive: Zone claim rate deviates $>2.5\sigma$ from baseline $\Rightarrow$ fraud
threshold auto-tightens + admin alerted — no human needed.

Every technology choice is justified by the architecture — not convention.

Layer	Technology	Why This Choice
Mobile App	Flutter (Dart)	Android-first; sensors_plus · barometer · razorpay_flutter
Auth + Realtime	Firebase	Phone OTP · real-time policy listeners · FCM push notifications
API + Logic	FastAPI (Python)	One language for API + ML — eliminates Node↔Python bridge overhead
ML / Fraud	XGBoost + scikit-learn + networkx	Premium pricing · payout scorer · Louvain graph clustering
Telemetry DB	MongoDB Atlas	Unstructured sensor payloads + fraud graph — Firestore limits would fail
Payments	Razorpay Payout API	Flutter SDK · UPI + IMPS · sandbox test mode for demo
Device Integrity	Google Play Integrity API	Detects rooted devices + mock GPS apps — free via Firebase
Admin Dashboard	React + Tailwind (Web)	Loss ratios · fraud radar · risk heatmap · predictive calendar

Market Opportunity

- ▶ **8M+** gig delivery workers in India
- ▶ **Rs27-54 Cr** annual uninsured loss, Bengaluru
- ▶ **2 Lakh** target subscribers — Year 1
- ▶ **58%** blended loss ratio — viable

Unfair Advantages

- ▶ Emergency 24-hr micro-coverage during alerts
- ▶ Zone Ambassador peer-verification network
- ▶ Predictive loss calendar for insurers
- ▶ Safe Week Streak — reduces moral hazard

Roadmap

Mar 20	Architecture · Persona · Fraud strategy · Premium model
Apr 4	Flutter app · Firebase Auth · FastAPI backend · Trigger engine
Apr 17	Full MSTS fraud engine · Admin dashboard · End-to-end demo
Year 1	Pilot: 500 Blinkit riders Bengaluru · Live product · 2L subscribers
Year 2-3	Break-even · Food delivery expansion · API platform · 5 cities

Key	Value
Persona	Q-commerce riders · Bengaluru
Event	Guidewire DEVTrails 2026
Premium	Rs15–Rs60 / week · weekly cycle
Payout	< 60 seconds · UPI / IMPS
Fraud defense	MSTS · 5 signals · Louvain graph
AI stack	XGBoost + scikit-learn + networkx
Mobile	Flutter + Firebase + FastAPI
Loss ratio	~58% blended · industry std

**“We don’t trust location.
We verify reality.”**

— Team Code Nexus

Parametric · Real-time · Fraud-resistant
Built for India’s 8M+ gig workers

Thank You

Nexurance AI

Trust the Person. Verify Reality.

Team Code Nexus · Guidewire DEVTrails 2026